



Good countries or good projects? Comparing macro and micro correlates of World Bank and Asian Development Bank project performance

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Published online: 5 September 2016

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Abstract This paper examines the micro and macro correlates of aid project outcomes in a sample of 3797 World Bank projects and 1322 Asian Development Bank projects. We find that project outcomes vary much more within countries than between countries: only 10–25 % of the variation in project outcomes is between countries, and the rest is within countries. Among country-level macro variables, GDP growth and the policy environment are significantly positively correlated with project outcomes. Among project-level micro variables, shorter project duration and the presence of additional financing are significantly correlated with better project outcomes. In addition, the track record of the project manager in delivering successful projects is highly significantly correlated with project outcomes. We find few significant differences between the two institutions in the relationship between these variables and project outcomes.

Keywords World Bank · Asian Development Bank · Project success

1 Introduction

A large literature has studied the effectiveness of development aid, with two broad areas of interest. The first strand has focused on the country-wide effects of aggregate aid inflows, typically on aggregate outcomes such as per capita GDP growth. Out of

Electronic supplementary material The online version of this article (doi:10.1007/s11558-016-9256-x) contains supplementary material, which is available to authorized users.

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necessity, this literature has emphasized the role of macro country characteristics, such as the overall policy and institutional environment, in intermediating the effects of aid on growth. The second strand in this literature has emphasized the micro side of aid effectiveness, based on the in-depth evaluation of specific aid-financed development interventions at the project level. Out of necessity, this literature has mostly emphasized the role of project characteristics in determining the effectiveness of specific interventions. However, much less is known about the *relative* importance of country versus project characteristics in driving aid effectiveness. Yet understanding the role of country versus project characteristics is of considerable importance to large aid donors that implement many projects in a broad cross-section of countries. For example, large multi-country donors need to decide how aid is allocated across countries, as well as within countries across specific projects, and they therefore also need to determine the role of country-level and project-level characteristics in their decision rules.

Systematic analysis of large numbers of projects implemented by multilateral development banks, and assessed using reasonably common standards, allows us to shed light on this important issue. In this paper, we build on earlier work by Denizer et al. (2013) (DKK), who investigate the macro and micro correlates of World Bank project outcomes using data from over 6000 World Bank projects between 1983 and 2011. Although DKK find that project outcomes are strongly correlated with country-level macro institutions and economic conditions, country-level factors account for only 20 % of the total variation in project outcomes. The remaining 80 % of the variation occurs within countries across projects. DKK investigated the relationship between a large set of project characteristics and project outcomes, and documented the role of factors such as project size, project length, the effort devoted to project preparation and supervision, and early-warning indicators that flag problematic projects during the implementation stage in explaining the within-country variation in project outcomes. DKK also documented a strong role for project manager effects in accounting for the variation in project outcomes. Geli et al. (2014) (GKN) extend these results to develop an empirical model for predicting eventual project evaluation outcomes, for the set of ‘active’ projects under implementation in the World Bank’s portfolio.

This paper extends the analysis of DKK and GKN by comparing the correlates of project success in the Asian Development Bank (ADB) with those in the World Bank (WB). This allows us to study similarities and differences across the two institutions in the relationship between project outcomes and country and project characteristics. To our knowledge, few studies have compared the organization-specific determinants of project success. One notable exception is Honig (2014), “Letting the Driver Steer: Organizational Autonomy and Country Context in Delivering Better Aid” unpublished, who examines the relationship between a particular country characteristic (fragility) and development organization characteristics (autonomy of local implementers and autonomy of the institution itself from political interference) in determining aid effectiveness, finding that more “autonomous” international development organizations are more successful operating in fragile states than organizations with less autonomy. Comparing correlates of project success between the WB and the ADB is interesting, given that the ADB works with a different set of clients, and with a different organizational culture from the WB. Despite these differences, a striking finding of this paper is that many of the relationships between project outcomes and country- and project-level variables documented in DKK for the WB extend also to the ADB.

Using data from 3797 World Bank projects and 1322 ADB projects, we document that project success rates vary more within countries than across countries. In the two institutions, country-level characteristics explain only 10–25 % of project success, indicating an important role for project-specific factors in understanding project outcomes. Among country-level factors, we find that GDP growth and a good policy environment are positively correlated with project success, consistent with DKK. In contrast with DKK, we find that across projects in Asia for both institutions, civil liberties and political freedom at the country level are negatively correlated with project outcomes. With regard to the micro correlates of project success, we find that projects that take longer to implement are less likely to be successful. We also find that the difference between actual and initially-planned funding is positively correlated with project outcomes, and we show that most of this is due to projects that are not doing well being closed early, as opposed to the scaling up of successful projects.

We also find that the track record of the project manager (known as the “task team leader” in the World Bank, and the “project officer” in the Asian Development Bank), defined as the success rate of the manager of a given project on other projects s/he has managed, is a very strong correlate of eventual project outcomes. As we discuss in more detail below, there is some element of self-evaluation in the project outcome ratings we study. Nevertheless, we document that our findings on the role of project manager track record in driving project outcomes remain significant after controlling for possible self-evaluation biases, an issue that was not considered in earlier work by DKK and GKN. We also find that turnover of project managers is negatively correlated with project outcomes in the WB and the ADB. Finally, leading indicators of project success, in the form of negative self-assessments by staff during the first half of a project, are also correlated with eventual project outcomes. In terms of differences across institutions, in most cases we cannot reject the null hypothesis that the magnitude of the relationship between these macro and micro correlates and project outcomes is the same across the two institutions.

Beyond its immediate antecedents in DKK and GKN, this paper contributes to a growing literature that has studied aid effectiveness by analyzing outcomes of public spending projects financed by multilateral development banks. An early contribution was the 1991 World Development Report (World Bank 1991) which noted higher economic rates of return on World Bank financed projects in countries with good policy performance, based on background research that was eventually published as Isham and Kaufmann (1999). Related work by Isham et al. (1997) documented the significance of political rights and civil liberties for rates of return in a global sample of World Bank financed projects. More recently, papers such as Dollar and Levin (2005) and Guillaumont and Laajaj (2006) consider other country-level factors such as policy quality and macroeconomic volatility in accounting for project-level success. Smets et al. (2013) showed that recipient country variables – namely, ideological orientation of the government – help predict loan effectiveness by influencing the level of effort that staff expend during the loan design phase.

As we have already noted, however, the vast majority of the variation in project outcomes occurs within countries across projects, indicating at best a limited role for country-level factors in accounting for project level success rates. The second part of our paper, which explores this, builds on several recent studies that have also examined project-level correlates of project outcomes, including Dollar and Svensson (2000);

Chauvet et al. (2015), and Kilby (2000, 2013, 2015). Relative to these studies, we emphasize (i) the distinction between cross-country versus within-country variation in project outcomes,¹ (ii) the role of project manager quality in driving project outcomes, (iii) the role of early warning signals of project outcomes coming from internal project monitoring processes, and (iv) the institutional comparison between the WB and the ADB.

The rest of this paper proceeds as follows. Section 2 describes the project outcome ratings for the WB and the ADB and provides some basic data description. Section 3 investigates the relationship between project outcomes and a variety of ‘macro’ country-level variables and ‘micro’ project-level variables, emphasizing comparisons between the WB and the ADB. Section 4 uses a smaller sample of projects for both institutions where we have information on the identity of the project manager of the project, to investigate the role of project management and the role of leading indicators in the form of negative outcome ratings during the first half of a project’s life. Section 5 discusses policy implications and concludes.

2 World Bank and Asian Development Bank project outcome ratings

We begin with a sample of 5038 WB projects exiting the World Bank’s portfolio between 1995 and 2012. This is the same set of projects used in GKN. For the WB as a whole, over 10,000 projects have been completed since the late 1940s. However, we focus on this more recent set of projects because of the more complete information on project characteristics available for them, most notably information on the identity of the project manager. For the ADB we begin with a sample of 1696 projects exiting the ADB’s portfolio between 1973 and 2012, as obtained from its evaluation databases. Eliminating a few outliers, and restricting the regression sample to projects with complete data for all the main explanatory variables of interest, limits the sample to 1322 ADB projects and 3797 World Bank projects, for a total of 5119 projects. The subsequent discussion and analysis refers to this restricted sample.

WB and ADB loan- and grant-financed activities are organized by projects. Most often these projects finance particular public sector activities, such as infrastructure projects, health and education initiatives, and a myriad of potential other development-oriented government actions supported by aid donors. In some cases projects simply take the form of budget support, subject to conditions that governments need to meet in order for disbursement to occur. To give a sense of the diversity of these projects, Table 1 reports the distribution of projects across sectors.² Compared with the WB, the ADB tends to have a larger share of projects in agriculture and in finance (even within Asia), but a smaller share of projects concentrated on public administration and health.

Projects are identified and designed through a collaborative process involving development bank staff and their counterparts in the country where the project will be implemented. A key ingredient of the WB project design process is the identification of the project’s “development objective,” which summarizes what the project is

¹ For related work on this issue in a very different context of aid projects in different communities in Pakistan, see Khwaja (2009).

² World Bank projects can be assigned to multiple sectors, with an indication of the fraction of the project’s value in each sector. We follow DKK in assigning each project to its largest sector. In the ADB data we have information on only one sector per project, which is reflected in the table above.

Table 1 Distribution of Projects across Sectors

	World Bank		World Bank-Asia		Asian Development Bank	
	# of projects	Project value	# of projects	Project value	# of projects	Project value
Agriculture	11.9 %	9.6 %	14.8 %	11.5 %	26.2 %	15.7 %
Transport	12.0 %	13.9 %	14.6 %	16.9 %	16.9 %	21.1 %
Public admin.	22.8 %	18.9 %	15.4 %	10.5 %	2.6 %	4.7 %
Energy	8.9 %	13.9 %	12.6 %	19.8 %	14.4 %	19.6 %
Education	10.4 %	7.8 %	10.1 %	7.5 %	8.2 %	4.9 %
Finance	5.7 %	10.5 %	5.7 %	10.3 %	9.2 %	14.2 %
Water	8.0 %	6.8 %	8.7 %	7.3 %	10.7 %	8.2 %
Industry	7.1 %	9.4 %	7.3 %	8.8 %	4.2 %	4.0 %
Health	12.0 %	8.3 %	9.4 %	6.5 %	3.3 %	2.6 %
Other	1.2 %	1.0	1.2 %	0.7 %	4.3 %	5.0 %
Total number/value (billion 2005 USD)	3797	418.5	1143	167.2	1322	127.6

This table presents the distribution of projects (both number of projects and the total value of projects by initial ADB/WB commitment) in the World Bank and ADB, as well as the distribution of World Bank projects in ADB countries. Project values are determined by initial commitments in constant 2005 USD

intended to achieve. Upon completion, projects are assessed according to their success in achieving this development objective. In addition, over the course of project implementation, project managers regularly report on the status of the project using the Implementation Status and Results Report (ISR). These ISRs include the project manager's assessment of whether the project is making good progress relative to its development objective (DO), using the same rating scale that is used to ultimately assess the project (as discussed below). We use these Implementation Status Report Development Objective (ISR-DO) ratings as an interim assessment of overall project quality. Analogously, project managers in the ADB regularly fill out interim Project Performance Reports and provide "Impact and Outcome" (IO) ratings, to predict the possible achievement of impact and outcome by completion date based on current assumptions and risks. Below we discuss in more detail these interim ratings and their usefulness in predicting eventual project outcomes.

Upon completion, WB projects are self-assessed by the project manager, in the form of an Implementation Completion Report (ICR). In the WB, during the post 1995 period that we consider, all Implementation Completion Reports were desk-reviewed by the WB's Independent Evaluation Group (IEG). The summary evaluation at this stage consists of a six-category rating of the project's outcome relative to its development objective (*Highly Unsatisfactory/Unsatisfactory/Moderately Unsatisfactory/Moderately Satisfactory/Satisfactory/Highly Satisfactory*). In addition, roughly 25 % of projects are subject to more detailed reviews based on field missions by the Independent Evaluation Group, which produces a detailed evaluation study of the project known as a "Project Performance Assessment Report." These reports also assess the project's outcome relative to its development objective, using the same six-category scale. We use these detailed evaluation ratings for all projects where they

are available, for a total of 1019 projects, and the ICR review-based evaluations for the remaining 2778 projects in our WB sample of projects. Finally, we convert the six point scale to a binary *Successful/Unsuccessful* classification by grouping the three gradations of each together.

The ADB follows a broadly similar process. Self-evaluations by project managers are known as Project Completion Reports (PCRs). They are done for all completed projects which incurred expenditures. Since 2007, PCRs have been desk reviewed by the ADB's Independent Evaluation Department using a Project Completion Validation Report (PVR). In addition, nearly 50 % of projects in our sample receive detailed evaluations in the form of Project Performance Evaluation Reports (PPER). Most of these were done prior to 2007 when the validation system started. As in the World Bank data, we use the most detailed evaluation available, i.e., the PPER if available, otherwise the PVR, and if neither is available we rely on the staff self-assessment in the form of the PCR. Finally, we note that prior to 1995, we have evaluations only for projects that were subject to a full PPER. Staff self-assessments through PCRs prior to 1995 were not formally rated, and are therefore not included in our sample for analysis.³

ADB projects evaluated before 2000 were rated on a three point scale (*Unsuccessful/Partly Successful/Generally Successful*). After 2000, the ADB split the *Generally Successful* category in two, resulting in a four point scale (*Unsuccessful/Less than Successful/Successful/Highly Successful*). To generate a binary success rating for use in our combined analysis of WB and ADB projects, we define the first two categories as *unsuccessful* in both the pre-2000 and post-2000 periods, and all *Generally Successful/Successful/Highly Successful* projects as *successful*. Figure 1 shows the distribution of projects across the various outcome categories, pooling all projects for the WB (left panel), and for the ADB (right panel).

One important feature of this data to keep in mind is that, while there are obvious similarities in the terminology used to define the rating scale for the two institutions (using gradations of “success”), the overall success rate of projects across the WB and ADB may not be fully comparable, as evaluators at the two institutions may have somewhat different standards and norms for determining what constitutes a “successful” (ADB) or “satisfactory” (WB) rating. For instance, the ADB includes a sub-rating for sustainability in the determination of the overall success rating; the WB has since the early 2000s produced a separate rating for sustainability, which they call ‘*risk to outcomes*.’ There are also differences in the distribution of projects across countries, which further complicates the comparison of overall aggregates of project success. In the empirical specifications that follow, we include a separate intercept for the WB and the ADB, which will pick up any differences in average success rates across the two institutions. Also, to pick up any differences in project success rates across both sectors and institutions, we include a full set of sector dummies for both institutions.

There are a variety of other reasonable concerns about the validity of these project outcome ratings. One general concern is that all projects are assessed relative to their development objective, rather than relative to any absolute standard. This introduces the possibility that at least some of the variation in project outcome ratings is due to

³ The lack of PCR ratings prior to 1995 explains the high share of projects (48 %) in our sample that receive detailed Project Performance Evaluation Reports (PPERs). From 1995 onwards, 292 out of 980 ADB projects have PPERs, for a rate of 30 % which is comparable to the WB.

Distribution of Project Outcome Ratings

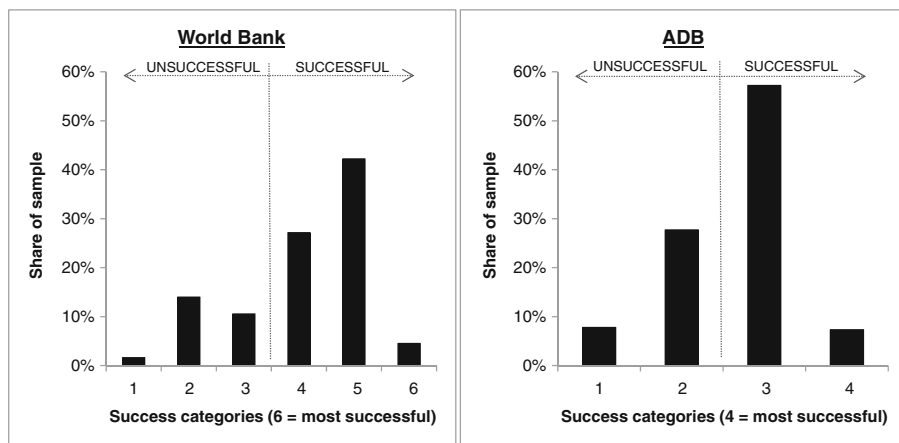


Fig. 1 Distribution of project outcome ratings. Notes: For ADB projects evaluated before 2000, a three point scale was used (“Unsuccessful,” “Partly successful,” and “Generally successful”). The first two categories match categories post-2000, but the “Generally successful” category splits into “Successful” and “Highly successful.” To generate this graph, we distributed the pre-2000 “Generally successful” ratings into the post-2000 successful categories (“Successful” and “Highly Successful”) assuming the same distribution between the two (88 % “Successful” and 12 % “Highly successful”)

differences in the ambition or attainability of the stated development objective, rather than due to any differences in actual outcomes. To some extent this problem is inevitable, given the wide variety of sectors in which the ADB and the WB operate: it would be difficult to imagine a common absolute standard that could be applied to the literally thousands of very different projects that these institutions have financed. This also means that differences in project success rates across sectors should not be taken too seriously, as they may reflect both differences in actual outcomes as well as differences in standards of setting project development objectives across sectors.

Another potential concern is that we are pooling results from two different types of evaluations for the WB, and three types for the ADB. For example, one might be concerned that project success ratings based primarily on the views of staff implementing the project (in the form of PCRs in the ADB, and their desk reviews in both institutions (the ICR and the PVR)) may be overly optimistic, as staff who are “close” to projects may naturally be more reticent about admitting less than satisfactory performance. To capture this possibility, in the regressions that follow we include two dummy variables for evaluation type. The first takes value equal to one if the project received a full evaluation in the form of a PPAR/PPER, and zero otherwise. The second, relevant only for ADB projects, takes value one for projects that receive PVRs, and zero otherwise. For the ADB, the omitted category consists of projects that receive only PCRs and are not desk reviewed or evaluated through a PPER (34 % of projects in the ADB sample).

A final caveat we should note is that these project evaluations are by no means well-identified (in the econometric sense) impact evaluations. Rather, they are reasonably careful administrative assessments which, in the case of PPARs and PPERs, generally also rely on special project field visits held within two years of project completion and careful documentation of the findings, with economic and

financial analysis done where feasible. For several major categories of projects, such as in finance or public administration, it would be difficult to conduct a rigorous impact evaluation; many projects are furthermore dispersed over many project sites and rigorous impact evaluations would then be difficult and costly to organize. With these caveats in mind, Fig. 2 shows trends over time in average performance across evaluation types in the World Bank and ADB. In the World Bank, average project success rates have been fairly stable between 1995 and 2015, fluctuating between 70 % and 80 %. Average project success rates have varied more within the ADB, but this is at least in part due to the fact that the number of projects included in each year's average success rate is considerably smaller than in the WB.

Figure 3 gives a first sense of the extent of variation in project outcomes across Asian countries in which both the WB and the ADB are active. The concentration of observations in the upper left hand quadrant reflects the higher success rate across World Bank projects, demonstrating that this higher success rate is not simply an artifact of different project distribution across countries for the two institutions. However, differences between the ADB and the WB in the distribution of projects across Asian countries does account for some of the difference in overall success rates shown in Fig. 3. For instance, in the sample 17 % of World Bank Asia projects are in China, which has a high project success rate for both institutions (90 %), while only 7 % of ADB projects are in China. If we assume ADB country success rates but apply World Bank Asia country distribution of projects, the ADB overall success rate would rise from 64.5 % to 70.9 %; similarly,

Trends in Average Project Performance by Evaluation Type and Institution

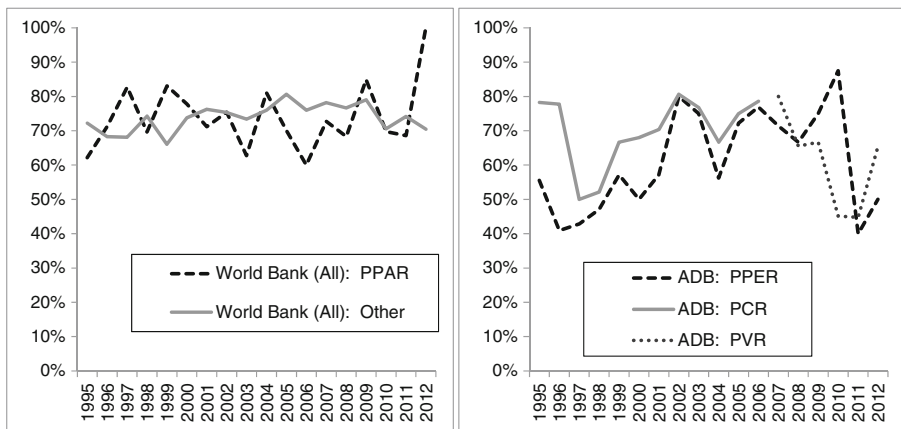


Fig. 2 Trends in average project performance by evaluation type and institution. Notes: These charts show average project success ratings by evaluation year and evaluation type. For the World Bank, PPAR refers to “Project Performance Audit Reports,” the more detailed project evaluations conducted by the Independent Evaluation Group. Note that 2012, the last year in our sample, is only partially complete and contains just 15 PPAR evaluations, all of which coincidentally gave successful ratings. In most previous years, we have 30–40 PPAR evaluations. For the ADB, PPER refers to detailed “Project Performance Evaluation Reports”; PCR refers to “Project Completion Reports” (self-evaluations by project managers); and PVR refers to “Project Completion Validation Reports” (desk reviews of PCRs). For the period after 2006, few PPERs have been done (about 10 a year), as the ADB started its PCR validation process in 2007. The PCR series is truncated in 2006 in the graph. However, the sample of projects used in the regression analysis contains a small number of unvalidated PCRs from 2007 onwards

applying ADB country distribution to World Bank country success rates lowers the World Bank Asia success rate from 78.2 % to 75.8 %.⁴

Although Fig. 3 shows that there is non-trivial variation across countries in average project performance for both institutions, a key feature of WB project ratings documented in DKK is that most of the variation in project outcomes occurs across projects within countries. Put differently, while there clearly are cross-country differences in average project success rates, as demonstrated in Fig. 3, there is also a great deal of variation within countries, with successful and unsuccessful projects coexisting in the same country.

One way to document this within-country variation directly is to consider a regression of project outcomes on country dummy variables. The R-squared from such a regression corresponds to the share of the variation in project outcomes in a given year that can be accounted for by differences in average performance across countries. Specifically, for each year from 1995 to 2007, we take the set of projects active in that year, and regress the eventual project outcome on a set of country dummies. We do this for WB and ADB projects separately, and also for WB projects implemented in the same set of countries as the ADB. The resulting R-squareds are quite low, varying between 10 and 25 %, and averaging 17.8 % (WB), 14.3 % (WB Asia), and 14.6 % (ADB) in the three samples.⁵ This motivates an analysis of the project-level correlates of success as distinct from country-level correlates, which we turn to next.

3 Correlates of project outcomes

We next document the empirical relationship between project outcomes and a variety of country-level and project-level characteristics.⁶ We do so using a series of probit regressions of the binary outcome rating on these variables, pooling all WB and ADB observations.⁷ We report results for a pooled sample of all projects from both institutions, and for a sample consisting only of WB and ADB projects in Asia.⁸ In the probit results, we report estimated marginal effects for all continuous explanatory variables, and changes in predicted probabilities for the discrete explanatory variables. In addition to documenting the characteristics of successful projects, we also are

⁴ Average project success rates vary across sectors, and the sectoral composition of projects differs slightly between the ADB and the WB. However, these differences are sufficiently small that they do not account for much of the difference in overall project success rates between the two institutions. Assuming ADB sector success rates and applying the WB Asia sector distribution results in a success rate of 64.0 % compared to the ADB baseline of 64.5 %. Similarly, assuming the WB sector success rates and applying ADB sector distribution results in a success rate of 77.3 % compared to a WB Asia baseline of 78.2 %. The different treatment of sustainability issues in evaluations in the World Bank and ADB in the 2000s may also explain part of the difference. Sustainability sub-ratings are generally somewhat lower than overall ratings in the ADB.

⁵ We cut off the analysis in 2007 due to the more limited number of projects to analyze from 2008 onwards, which artificially drives the R-squareds higher.

⁶ Summary statistics on all variables used in the empirical analysis can be found in Appendix Table 1.

⁷ We obtain broadly similar results if we use a linear probability model (as in DKK), or if we estimate an ordered probit model separately for the WB and the ADB, so as to use the information in the 1–6 scale of the WB and the 1–4 scale of the ADB ratings.

⁸ As a further robustness check, we also estimated our specification for a sample that excluded ADB projects evaluated prior to 1995, in order to match the timing of the WB project sample. However, the results are very similar in this smaller sample, and so are not reported to conserve space.

Distribution of Average Success Rates by Country

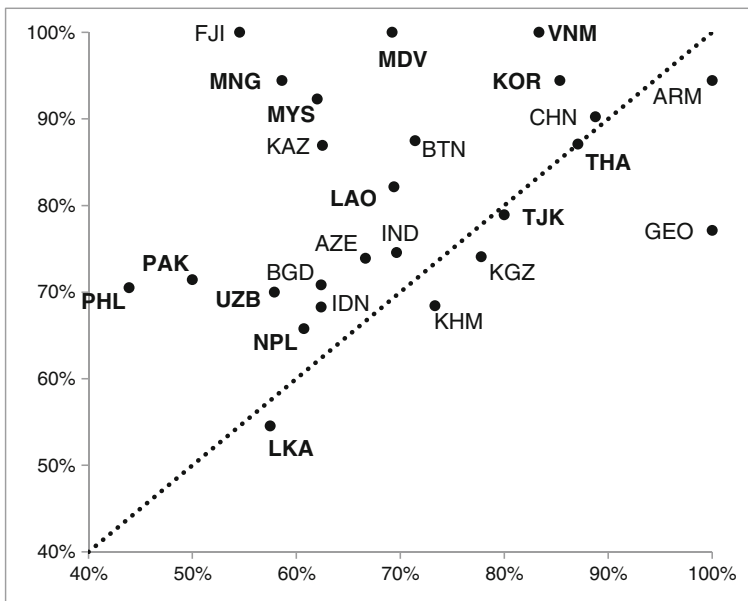


Fig. 3 Distribution of average success rates by country. Notes: This chart compares average project success rates by country in the World Bank (y axis) and ADB (x axis). All countries with significant differences in success rates between the two institutions (indicated by z scores greater than two) are identified by bolded and underlined country labels. Observations with labels in gray do not have significantly different mean success rates across the two institutions. The ISO country codes correspond to: Armenia (ARM), Azerbaijan (AZE), Bangladesh (BGD), Bhutan (BTN), Cambodia (KHM), China (CHN), Fiji Islands (FJI), Georgia (GEO), India (IND), Indonesia (IDN), Kazakhstan (KAZ), Kyrgyz Republic (KGZ), Lao People's Democratic Republic (LAO), Malaysia (MYS), Maldives (MDV), Mongolia (MNG), Nepal (NPL), Pakistan (PAK), Philippines (PHL), Republic of Korea (KOR), Sri Lanka (LKA), Tajikistan (TJK), Thailand (THA), Uzbekistan (UZB), and Vietnam (VNM)

interested in the extent that these differ between the WB and the ADB. In order to facilitate this, the regressions include an interaction of each right-hand-side variable with a dummy variable indicating ADB projects, so that the interaction term picks up differences across institutions in the estimated relationship between the variable of interest and project outcomes.⁹

As discussed above, differences in the rating scales used by the WB and ADB to evaluate projects imply that there may also be differences in reported overall average success rates across the two institutions. In addition, we have discussed how project success rates might differ across sectors due to differences in the types of development objectives set for projects in different sectors. To pick up these differences, we include a full set of sector dummy variables, and their interaction with a dummy for ADB projects, in all the regressions that follow (not reported for reasons of space). All specifications also include evaluation year fixed effects and their interaction with the

⁹ Note that since every variable in the regression, including the unreported sector and period dummies, enters both directly and as an interaction with a dummy for ADB projects, this means that our specification is equivalent to estimating separate regressions for the two institutions. We estimate the pooled regression with interactions only to facilitate testing the null hypothesis of equal slopes across the two institutions.

ADB dummy, to pick up potential changes over time in evaluation standards in the two institutions (also not reported for reasons of space). To allow for differences in average success rates across evaluation types, all specifications include a dummy variable for projects where the outcome rating is based on the more detailed PPAR/PPER evaluation, as well as its interaction with an ADB dummy, and also a dummy variable for projects where the outcome rating is based on PVR evaluations (relevant only for ADB projects) to pick up any effects of evaluation type. Finally, all specifications include a dummy variable for whether the project is investment lending (i.e., lending to finance a specific project) versus policy lending (i.e., budget support not directly tied to a specific project), as well as its interaction with an ADB dummy.

In the first two columns of Table 2 we consider the relationship between three major country characteristics and project outcomes. Since most projects require several years to implement, we measure each of these country characteristics as the annual average of the variable in the country and over the years in which the project was implemented. The first is a measure of country-level policy performance, the Country Policy and Institutional Assessment (CPIA) ratings of the World Bank.¹⁰ The CPIA rates countries on 16 criteria in four clusters: economic management, structural policies, policies for social inclusion and equity, and public sector management. CPIA ratings are very strongly positively and significantly correlated with project outcomes across all specifications. The magnitude of this effect also is substantial. A one-point movement (on the six-point CPIA scale) is associated with a 14 % higher probability of a successful project outcome rating. This intuitive relationship supports the findings of Dollar and Levin (2005), who demonstrate that World Bank project success rates in the 1990s increase with institutional quality in recipient countries. Similarly, Isham and Kaufmann (1999) show that the economic rate of return on public and private investments increases within countries as economic policymaking improves. The same relationship between better country-level policy performance and World Bank project outcomes is documented in DKK and GKN.

We also consider country-level real GDP growth rates over the period in which the project was implemented. Echoing the finding in DKK, projects implemented in fast-growing countries are significantly more likely to be rated as successful. An additional percentage point of average growth over a project's life is associated with a probability of project success that is around 1.7 % greater at the margin.

Finally among the country-level variables, we consider the index of civil liberties and political rights produced by Freedom House. We use this particular measure to be consistent with earlier work that has related this country-level variable to project-level success rates, and also for the pragmatic reason that it is the only such measure available for the full time span and set of countries included in our dataset. Freedom House scores both of these indicators on a 1–7 scale. We sum them together and reorient to arrive at a scale from 0 to 12, with higher values corresponding to higher civil liberties and political rights. In the global sample and for WB projects, there is no significant correlation between this measure and project outcomes. However, in the Asian sample the

¹⁰ The ADB produces a similar set of assessments, but only for poorer client countries eligible for concessional lending from the Asian Development Fund. In contrast, the WB produces its CPIA assessments for all clients, but discloses them publicly only for concessional borrowers. For concessional borrowers in Asia, the WB and ADB CPIA ratings are quite highly correlated.

Table 2 Correlates of Project Outcomes

Dependent variable is binary success (0,1) in all specifications

Explanatory variable:	(1)	(2)	(3)	(4)
	Full	Asia	Full	Asia
CPIA rating	0.144*** (6.51)	0.140*** (3.35)	0.140*** (6.16)	0.130*** (3.08)
—ADB interaction	−0.0533 (−1.15)	−0.0500 (−0.96)	−0.0443 (−1.11)	−0.0347 (−0.62)
Real GDP per capita growth	1.776*** (5.31)	1.583* (1.86)	1.685*** (5.05)	1.080 (1.60)
—ADB interaction	0.339 (0.32)	0.512 (0.41)	−0.647 (−0.76)	−0.0486 (0.05)
Freedom House rating	0.00149 (0.35)	−0.0115* (−1.70)	0.00200 (0.48)	−0.00952 (−1.59)
—ADB interaction	−0.00996 (−1.38)	0.00310 (0.41)	0.00660 (−0.93)	0.00495 (0.79)
Dummy for PAR/PPER evaluations (d)	−0.00576 (−0.30)	0.00678 (0.22)	−0.0284 (−1.47)	−0.00585 (−0.19)
—ADB interaction (d)	−0.0814** (−2.06)	−0.0927** (−2.14)	−0.0829* (−1.89)	−0.105** (−2.07)
Dummy for ADB PVR evaluations (d)	−0.208*** (−3.48)	−0.205*** (−3.43)	−0.195*** (−3.26)	−0.192*** (−3.25)
Dummy for investment lending instrument (d)	−0.0882*** (−4.24)	−0.0824 (−1.63)	0.0896** (2.25)	0.0693 (0.74)
—ADB interaction	0.133*** (3.64)	0.136*** (2.81)	0.0781 (1.53)	0.0990 (1.13)
Log(total commitment)			0.0239*** (2.95)	0.0132 (0.71)
—ADB interaction			−0.0131 (−0.98)	−0.00249 (−0.13)
Planned project length			−0.00219*** (−3.95)	−0.00208 (−1.59)
—ADB interaction			0.00151 (1.38)	0.00140 (0.88)
Effectiveness delay			0.00340 (1.57)	0.0108* (1.88)
—ADB interaction			−0.00143 (−0.45)	−0.00887 (−1.29)
Implementation delay			−0.00176*** (−3.45)	−0.000738 (−0.66)
—ADB interaction			−0.00156* (−1.93)	−0.00255** (−1.99)
Log(additional funding)			0.331***	0.340***

Table 2 (continued)

Dependent variable is binary success (0,1) in all specifications

Explanatory variable:	(1) Full	(2) Asia	(3) Full	(4) Asia
—ADB interaction			(10.13) −0.0368 (−0.60)	(4.39) −0.0477 (−0.45)
Observations	5119	2465	5119	2465

Table reports marginal effects from probit regression. “(d)” indicates discrete change of dummy variable from 0 to 1. T statistics based on heteroskedasticity-consistent standard errors clustered at the country level are reported in parentheses

*** (**) (*) denote significance at the 1 (5) (10) percent level. All regressions include sector and evaluation year fixed effects and their interactions with the ADB dummy

relationship is negative and marginally significant, with better project outcomes in countries rated by Freedom House as having fewer civil liberties and political rights. This differs from earlier findings. DKK find an insignificant (but positive) relationship between Freedom House ratings and project success, the same as our global sample. Isham et al. (1997) find a positive and significant correlation between civil liberties and the performance of government investment projects, using a global sample. Finally, it is noteworthy that in all cases, there is no evidence of a differential relationship between the macro variable of interest and project outcomes in the ADB as compared with the WB.

In the next two columns, we consider the relationship between a set of basic project characteristics and project outcomes. The first two are measures of planned project size, as proxied by the logarithm of the total commitment, measured in constant 2005 US dollars, and by the initial planned length of the project, measured in months from project approval to planned completion date. We find that planned project size is positively correlated with project success, and significantly so in the full sample, while planned length is negatively correlated with success in both samples, and significantly so in the full sample. The positive correlation with planned size is somewhat surprising, as it is the opposite of the findings in DKK. A possible interpretation is that projects with greater initial commitments are given greater attention; this effect may outweigh any complexity effect (i.e., projects with larger initial commitments are likely to be more complex and thus less likely to be successful). In the ADB case, China, Vietnam, and India tend to receive large loans, and these countries, particularly China and Vietnam, also have higher project implementation capacity and greater control over local factors such as land and local government. The negative relationship between project success and planned length is more intuitive: projects that are expected to take longer to implement are more likely to be more complex, and therefore also have a greater risk of receiving unsuccessful ratings.¹¹ The magnitude of this effect is non-trivial — an additional year of planned project length is associated with a 2.4 percentage point lower probability of a successful outcome rating.

¹¹ It is important to emphasize that we are looking at *planned* project length, not *actual* project length. Complex projects may get restructured or cut off and then rated unsuccessful. As components get canceled, the actual project length may be quite short.

The next two indicators capture delays in the process of project implementation. The first, effectiveness delay, measures the time (in months) between project approval and project “effectiveness,” i.e., the time from signing the loan to the time that all conditions of the loan agreement are declared fulfilled so that disbursements can be made. The second, implementation delay, measures the difference (in months) between the actual completion date of the project and the planned completion date. We find some weak evidence that longer “effectiveness” delays signal better project outcomes. A potential channel for this effect is that in some countries, experienced executing agencies wait to fulfill all conditions until detailed project designs have been completed and all procurement packages are prepared; this then reduces later delays after effectiveness for which special “commitment” charges (to discourage project startup delays) may be levied by the lender. The observed “delay” to declaration of effectiveness therefore indicates special care and attention (which then enhances the speed and quality of subsequent implementation). Intuitively, we find evidence that projects that take longer than expected after loan effectiveness to complete are more likely to be rated poorly, as these implementation delays are likely to proxy for factors undermining project success. This effect is statistically significant in the full sample.

We also include a variable measuring the difference between the (log) final disbursements and (log) initial commitment on the project, i.e., the percentage difference between the total amount eventually disbursed on the project, and the initial commitment. These differences can arise for a variety of reasons. In some cases, total disbursements are less than the initial commitment if the project was performing poorly and is closed early or scaled down. Conversely, projects that are performing well may receive additional financing beyond their initial commitment. Together, these factors suggest we should find a positive relationship between this variable and project outcomes. On the other hand, sometimes additional financing is required to complete a project due to unforeseen cost overruns. In this case the relationship between this variable and project outcomes is less clear, since the effect of cost overruns on the eventual quality of the project is unclear.

We find that this disbursement/commitment gap (which for convenience we refer to as “additional financing”) is strongly and significantly correlated with project success, lending support to the first interpretation. The magnitude of this effect is large as well. A project that receives 20 % (18 log points) less funding than originally planned has a six percentage point lower chance of receiving a successful rating. Interestingly, most of the explanatory power of this variable comes from cases where disbursements are less than initial commitments, i.e., where disbursements were smaller than initial commitments. If we set all positive disbursement gaps to zero, the estimated effect of this variable remains strongly significant. Conversely, if we set all negative disbursement gaps to zero, the estimated coefficient falls by half and is substantially less significant.¹² Overall, this suggests that most of the relationship between project outcomes and the difference between initially planned commitments and eventual disbursements is mostly driven by unsuccessful projects being scaled back, and to a

¹² A potential concern here is that some of the difference between disbursements and commitments could reflect exchange rate fluctuations on loans denominated in currencies other than US dollars, that then are converted back to US dollars for recording purposes. To guard against this possibility, we also repeated our results, but dropping all log differences between disbursements and commitments that were less than 25 % in absolute value, so that the remaining “large” differences are more likely to be due to deliberate choices to scale down or scale up the project. Doing so has little effect on our results.

lesser extent to successful projects being expanded. Interestingly, none of the interactions between all of these project-level variables and the ADB dummy are significant, indicating that the magnitude of the relationship between these project characteristics and project outcomes is not significantly different between the WB and the ADB.

Finally, we note that in all specifications, the PPAR/PPER evaluation type dummy variable's interaction with the ADB dummy is negative and significant. In other words, in the ADB case, evaluation ratings based on more detailed PPERs and PVRs on average result in project ratings that are significantly lower than those based on Project Completion Reports, i.e., the omitted category. We also find that the ADB PVR evaluation type dummy variable is significantly negative, indicating that desk reviews of self-assessments provided by staff in Project Completion Reports in the ADB on average result in significantly lower scores. For the WB, there is no significant difference in project ratings coming from the two evaluation types (PPARs and ICR reviews). This is not surprising given that both of these are assessments produced by the WB's Independent Evaluation Group, while in the ADB sample the comparison is between different types of independent evaluations and the omitted category of self-evaluations in the form of Project Completion Reports. In terms of project type, we find evidence that investment lending projects get better average outcome ratings — the estimated coefficient in Column (3) of Table 2 indicates that these projects are associated with a 9 percentage point higher chance of a successful outcome rating. The effect is somewhat smaller and less significant in the Asia-only sample in Column (4).¹³

A challenge in interpreting all of these findings is that both project ratings and observed project-level variables to some extent respond to unobserved project characteristics that ultimately determine the success or failure of the project. For instance, as indicated above, projects that seem to be failing are likely to be cut short; may be more likely to receive additional funding in order to achieve success (or alternatively receive less money if such funds are seen as throwing good money after bad); and may receive less attention and are unable to finish on time. For these reasons, the partial correlations between project outcomes and project characteristics should be interpreted with some caution. For a more detailed discussion of the extent to which a causal interpretation can be assigned to these relationships, see Section 6 of DKK.

4 Project managers, interim indicators, and project outcomes

Thus far, we have discussed the project and country-level correlates of project success, noting that the relationship between these factors and project outcomes is broadly similar for the WB and the ADB. In this section, we turn to the role of project manager

¹³ Interestingly, investment lending projects tend to receive lower ratings in Columns (1) and (2) which do not include project-level controls – indicating that investment lending projects tend to have other characteristics such as longer planned length and longer implementation delays that are associated with worse performance. Once these attributes are held fixed, investment lending projects do better on average. Note that this sign reversal holds true only for WB projects. In Columns (1) and (2) the ADB interaction enters significantly positively, and with an estimated coefficient that is larger in absolute value than the direct effect. This means that, even without controlling for other project characteristics, investment lending projects do not receive lower ratings on average in the ADB.

experience and project manager turnover in driving project outcomes, as well as the information content of interim indicators of project outcomes produced as part of the process for monitoring projects during implementation. As background, recall that at every point in time, each project in our sample is the responsibility of a project manager (“Task Team Leader” in the WB, and “Project Officer” in the ADB). As project managers move around from one job to another within their respective institutions, responsibility for a given project often passes from one project manager to another. In this section, we develop measures of project managers’ experience in terms of overseeing projects that earn satisfactory ratings, as well as measures of project manager turnover for each project, and relate both to project outcomes.

Specifically, for each project, we calculate a project manager “track record” variable that reflects the weighted average success rate of all the other projects that the project manager has worked on, excluding the project in question, with weights proportional to the amount of time the project manager worked on each project. To see how this works, consider a hypothetical project manager who has managed projects A, B, and C, and we want to calculate the “track record” of the project manager for project C. We calculate the track record as a weighted average of the success rating of projects A and B, with weights proportional to the fraction of each project’s life that the project manager was responsible for those two projects. For the ADB sample, we only have data on project managers from 2000 onwards, limiting the sample and also weighting the project manager track record variable towards more recent years. In other words, if a project runs from 1995 to 2005, the track record variable for the ADB only reflects the project managers from 2000 to 2005 and their average project success on other projects from 2000 onwards. For both institutions, adding project manager track record limits our analysis to projects that have managers who have also worked on at least one other project in our sample, significantly reducing the sample size.

The following regressions also include a measure of the frequency of project manager turnover. We measure this as the number of managers per project-year over the life of a project. For ADB projects where we have project manager data starting only in 2000, for projects that begin prior to 2000 we use only the number of managers and the number of project - years from 2000 onwards, so as not to artificially bias downwards our measure of turnover for these projects. In the sample, managers of ADB projects turn over considerably more frequently than for World Bank projects: on average, ADB projects have 0.64 managers per project - year, while World Bank projects average 0.43 managers per project-year. Equivalently, ADB project managers serve an average of 1.9 years on a project, while WB project managers on average serve 2.9 years on a project.¹⁴ Including the managers per project - year variable and its interaction with the ADB dummy variable enables an analysis of the effects of project manager turnover in both World Bank and ADB projects.

In this section, we also consider the relationship between an interim indicator of project success and ultimate project outcomes. In both the WB and the ADB, project managers are required to periodically report on the implementation status of the

¹⁴ Note that the figures in this sentence are not the inverses of the figures in the previous sentence. This is because the average across projects of years per project manager is not the same as the average across projects of project managers per year. Note also that our figure of average tenure of 1.9 years for the ADB, derived from 10 years of interim progress report data from 2000 onwards, is similar to 1.7 years found by an analysis of staff tenure data of 375 PVRs (IED 2015). This suggests that the limited availability of pre-2000 data in our sample does not seriously affect our estimate of average turnover.

projects they are responsible for, typically twice per year. One element of this reporting is a self-assessment of the likely eventual outcome of the project. In the WB, these interim ratings are the ISR-DO (Implementation Status Report-Development Outcome) ratings. In the ADB they correspond to IO (Impact and Outcome) ratings, referring to the expected achievement of impact and outcome by the project completion date. Similar to the project manager data, we only have ADB IO rating data from 2000 onwards, and can thus only identify a negative rating in the first half of a project's life if the project's midpoint is after 2000. This limits us to 367 more recent ADB projects. For both institutions, the variable takes the value one when a negative rating is reported at any time during the first half of the project. For the ADB, this "negative" rating corresponds to an "*Unsatisfactory*" or "*Partly Satisfactory*" rating as opposed to a "*Satisfactory*" or "*Highly Satisfactory*" rating. This occurs in 8.6 % of ADB projects, versus 12.9 % of WB projects.¹⁵ A striking feature of these interim assessments is that they are quite optimistic –upon completion a significantly larger proportion of these projects are rated as unsatisfactory (31.1 % and 26.2 % in the ADB and World Bank, respectively). This likely reflects not only project quality deteriorating in the second half of implementation, but also excessive optimism about ultimate project outcomes on the part of project managers rating their own projects, and the instinct of some to underplay project problems as they may be associated by supervisors with the project manager's own performance.¹⁶

Our data on project managers and interim indicators is less complete than the data on other project characteristics. Restricting the sample to those projects for which the project manager track record, managers per project - year, and warning rating variables are all available leaves us with 2740 observations, consisting of 2373 World Bank projects and 367 ADB projects.¹⁷ The probit results reported in Table 3 below extend the specifications in Table 2, using this more limited sample. To ensure that the limited sample includes projects similar to the full sample, the first two columns of Table 3 repeat the baseline specifications from Table 2, in the reduced sample. Comparing the first two columns of Table 3 to columns 3 and 4 in Table 2, it is clear that restricting attention to the limited sample does not materially change the main findings discussed above, although some coefficients are less significant, as would be expected given the considerably smaller sample size. Columns 3 and 4 of Table 3 add the project manager track record variable, project manager turnover, and the first half rating variable, along with the interaction of all three variables with an ADB dummy. Column 3 corresponds to the full range of observations, while Column 4 reflects only World Bank and ADB projects in countries in which the ADB operates, as in Table 2.

¹⁵ This difference should not be over-interpreted as a sign of greater candor on the part of WB project managers, as it simply is an unconditional probability. Controlling for the same set of project characteristics included in the regressions in Table 2, we do not find that the rate of unsatisfactory self-assessments in the first half of project life is significantly different between the two institutions.

¹⁶ Optimism bias is a well-known feature of project management (Flyvbjerg 2006; Siemiatycki 2010; Kolkma 1999).

¹⁷ All World Bank projects in our sample have first half ratings. Including the warning ratings reduces the ADB sample significantly while including the project manager track record variable reduces the sample size for both institutions. The regression results reported below focus only on a sample with observations for both variables. Running separate regressions for each increases the number of observations but does not significantly alter the results.

In both the full and Asia sample, the coefficient on project manager track record is strongly significant, with no significant difference in the size of the effect between the World Bank and ADB. Note that the coefficient estimates are not just significant, but they are large as well. The estimated marginal effects imply that an increase in project manager track record by one standard deviation (0.28) is associated with a 6 percentage point higher chance of a successful outcome rating. In comparison, a one standard deviation (0.44) increase in the mean CPIA score is associated with a 4 percentage point higher chance of a successful outcome rating. This result is similar to that in DKK, who find that project manager track record and recipient country policy quality have comparably large partial correlations with project outcomes. Similar patterns hold in the Asia-only sample: a one standard deviation increase in project manager track record (0.27) is associated with a 7 percentage point higher success rating, while a one standard deviation (0.34) increase in the CPIA score is associated with a 2 percentage point higher success rating.¹⁸

The results for project manager turnover and interim project quality indicators also are similar to those in DKK. For both institutions, we find that higher project manager turnover is negatively and significantly correlated with project outcomes. We also find that negative interim project outcome ratings observed in the first half of project life are strongly correlated with unsatisfactory final project ratings. The coefficient estimates imply that a negative rating in the first half of project life is associated with a 70 percentage point lower chance of a successful outcome rating in both samples. Interestingly, this effect is somewhat smaller in the ADB — the ADB interaction is significantly positive, suggesting that in the ADB, a negative rating in the first half of project life reduces the chance of a satisfactory final rating by 50 percentage points. This difference between the WB and the ADB is however somewhat difficult to interpret. On the one hand, it could reflect a higher success rate in the ADB of interventions to “turn around” projects that are found to be performing poorly in the first half of their implementation period. On the other hand, it may reflect a tendency among project managers in the ADB to be more likely to acknowledge problems during project implementation only if those problems are easily remedied. For example, if problems are reported in the ADB, they need to be followed by special action plans. If the problem is fundamental and unlikely to be easily resolved, it may be less likely to be reported in the first place. Overall, however, there is a considerable degree of persistence within projects over time — in both institutions, there is a very high risk that projects flagged as problematic in their first half will ultimately turn out to have unsatisfactory final outcome ratings.¹⁹

One potential problem with the interpretation of project manager track record effects on project success is the possibility of systematic bias introduced by the element of self-evaluation in the project outcome ratings. As discussed in Section 2, many of the project evaluations in our sample are based on desk reviews by the WB and ADB

¹⁸ A potential concern here is that, despite the presence of country-level control variables in the regression, some project managers tend to work in “difficult” countries, and this in turn lowers the measured track record of the project manager. This however turns out not to be a factor — the project manager track record variable remains strongly significant even when country fixed effects are included in the regressions.

¹⁹ It is worth noting that including the first half interim project outcome ratings helps to control for some of the omitted variable bias problems that clouded the interpretation of other project-level correlates discussed above. This is because this indicator captures information observable to the project manager at the time about likely deficiencies in the project that are hard to measure *ex post*. The fact that coefficients on most of the other project-level variables retain their size and significance is therefore encouraging that these potential omitted-variable biases may not be too severe.

Table 3 Correlates of Project Outcomes Including Project Manager Records and “Warning” Ratings

Dependent variable is binary success (0,1) in all specifications

Explanatory variables:	(1)	(2)	(3)	(4)
	Full	Asia	Full	Asia
CPIA rating	0.152*** (5.19)	0.123** (2.24)	0.0957*** (3.23)	0.0412 (0.79)
—ADB interaction	−0.0433 (−0.42)	−0.0237 (−0.21)	0.0159 (0.17)	0.0564 (0.56)
Real GDP per capita growth	1.817*** (4.61)	1.184* (1.88)	1.061*** (2.74)	0.583 (1.17)
—ADB interaction	1.401 (0.74)	1.736 (1.01)	1.298 (0.79)	1.480 (1.03)
Freedom House rating	−0.00131 (−0.28)	−0.00919 (−1.44)	−0.00617 (−1.45)	−0.0101 (−1.64)
—ADB interaction	−0.00760 (−0.55)	0.00111 (0.10)	−0.00605 (−0.48)	−0.000623 (−0.05)
Dummy for PAR/PPER evaluations (d)	−0.0157 (−0.66)	0.0181 (0.72)	−0.0590** (−2.40)	−0.0303 (−0.85)
—ADB interaction (d)	−0.349*** (−2.86)	−0.380*** (−3.05)	−0.344*** (−2.77)	−0.353*** (−2.79)
Dummy for ADB PVR evaluations (d)	−0.316*** (−2.86)	−0.286*** (−2.78)	−0.335*** (−2.92)	−0.295*** (−2.84)
Dummy for investment lending instrument (d)	0.0340 (0.65)	0.0950 (0.87)	0.0606 (1.19)	0.147 (1.47)
—ADB interaction	0.103 (1.18)	0.0466 (0.37)	0.0569 (0.57)	−0.0236 (−0.18)
Log(total commitment)	0.0224** (2.43)	0.00939 (0.45)	0.0204** (2.45)	0.0302** (2.08)
—ADB interaction	0.00362 (0.11)	0.0142 (0.37)	−0.000994 (−0.03)	−0.0133 (−0.45)
Planned project length	−0.00169*** (−2.65)	−0.00173 (−1.48)	−0.00207*** (−3.04)	−0.00318*** (−3.45)
—ADB interaction	0.00133 (0.69)	0.00140 (0.72)	0.00112 (0.55)	0.00235 (1.16)
Effectiveness delay	0.00155 (0.57)	0.0128* (1.85)	0.00172 (0.61)	0.0104 (1.29)
—ADB interaction	−0.00662 (−1.26)	−0.0174** (−2.11)	−0.00609 (−1.14)	−0.0142 (−1.54)
Implementation delay	−0.00120* (−1.87)	0.000648 (0.45)	−0.00213*** (−3.33)	−0.00259** (−2.15)
—ADB interaction	0.00128 (0.71)	−0.000577 (−0.25)	0.00145 (0.82)	0.00199 (0.90)
Log(additional funding)	0.315***	0.296***	0.199***	0.166***

Table 3 (continued)

Dependent variable is binary success (0,1) in all specifications

Explanatory variables:	(1)	(2)	(3)	(4)
	Full	Asia	Full	Asia
	(7.74)	(3.43)	(6.51)	(2.75)
—ADB interaction	0.0744	0.0576	0.167**	0.154*
	(0.91)	(0.46)	(2.39)	(1.75)
Project manager track record			0.224***	0.252***
			(5.10)	(3.52)
—ADB interaction			0.0857	0.0186
			(0.82)	(0.17)
Project manager turnover			−0.113**	−0.173*
			(−2.01)	(−1.90)
—ADB interaction			−0.00731	0.0671
			(−0.07)	(0.57)
Negative rating in first half (d)			−0.700***	−0.717***
			(−28.04)	(−15.89)
—ADB interaction (d)			0.230***	0.197***
			(20.85)	(11.31)
Observations	2740	1122	2740	1122

Table reports marginal effects from probit regression. “(d)” indicates discrete change of dummy variable from 0 to 1. T statistics based on heteroskedasticity-consistent standard errors clustered at the country level are reported in parentheses

*** (**) (*) denote significance at the 1 (5) (10) percent level. All regressions include sector and evaluation year fixed effects and their interactions with the ADB dummy

evaluation departments of self-assessments done by project managers upon completion of the project. Moreover, our ADB sample includes some projects between 1995 and 2007 where the only rating we have is the staff-provided Project Completion Report (PCR) rating. If some project managers are systematically over- or under-optimistic in providing these assessments, and if this persists through the desk review process, then this may contribute to a spurious correlation of project manager performance across projects. We address this concern in two different ways. First, we exclude from a project manager’s track record past projects where s/he was the last manager, so that self-evaluations by that manager are excluded from his/her track record variable (Alternative #1). Second, we exclude the last manager of the project from the project-level measure of management track records (Alternative #2).²⁰

²⁰ Specifically, if w_i^j is the percent of the duration of project j for which i was the manager and PR_j is the rating for project j , then our default measure of the track record of project manager i is: $R_i = \sum_j w_i^j * PR_j / \sum_j w_i^j$ and the track record of the project J management team is: $R^J = \sum_i w_i^J * R_i$. Alternative #1 excludes from R_i all projects for which i was the last manager. Alternative #2 excludes the last project manager from the calculation of R^J .

Table 4 repeats the analysis of Table 3, but using these two alternative project manager track record variables instead. The first two columns repeat the baseline approach from Table 3 for the full and Asia-only sample, including the original project manager quality variable, but in the reduced sample of projects where we can compute the two alternative project manager track record variables. Columns (3) and (4) use Alternative #1 to calculate project manager quality, and columns (5) and (6) use Alternative #2. Consider first the findings in the full sample (Columns 1, 3, and 5). In the first column, we show that the baseline project manager track record variable remains strongly significantly correlated with project outcomes even in this smaller sample. When we move to Alternative # 1, which excludes from the project manager's record the projects s/he was the final manager of, we find a smaller, but still strongly significant correlation with project outcomes. Using Alternative # 2, we find a similar magnitude of effect as in the baseline measure, and also similar significance. These findings suggest that while there may be some self-evaluation bias, it is not sufficiently strong to overturn our conclusion that there is a strong and significant correlation between project manager track record and project outcomes. The evidence is more mixed when we turn to the much smaller Asia-only sample, which now consists of only 851 projects (Columns 2, 4, and 6). The baseline project manager track record variable still enters positively, although no longer significantly so — most likely due to the much smaller sample size. As in the full sample, the estimated coefficient is half as large when we use Alternative # 1, and is similar in size and significance to the benchmark when we use Alternative # 2. Overall, our findings here suggest that while project manager track record does matter for project outcomes, the size of the effect is smaller and less precisely estimated when we explore alternatives that are less likely to be affected by self-evaluation biases.

We conclude this section by probing further into the relationship between project manager turnover, interim ratings, and final project outcome ratings. As we saw in Table 3, in both the ADB and the WB, project manager turnover is negatively associated with project outcomes. This effect could in principle reflect the balance of two opposing forces. On the one hand, one might naturally expect that when a project is passed from one manager to another, some project manager-specific human capital is lost and project outcomes consequently suffer. On the other hand, if turnover is the result of assigning “better” managers to projects that are experiencing difficulty during implementation, then project manager turnover could also be associated with better outcomes, to the extent that managers with a better track record are more able to turn problem projects around.²¹ While our results are consistent with an overall negative effect of turnover on outcomes, it is nevertheless useful to try to document whether the second more positive channel is operative.

We investigate this by examining project manager “quality profiles” over project life-cycles. Specifically, we compute the change in project manager track record between the first half of the project and the second half of the project. We then calculate

²¹ While such turnover can be beneficial for the project in question, it may not be beneficial for the entire portfolio of projects the ADB or the WB are implementing. This aggregate effect will depend on both (a) what happens to the other projects the incoming project manager is also working on – it could be that the additional claims of the new project on the project manager's attention come at the expense of the other projects in his or her portfolio, and (b) what happens to the outgoing project manager, who may instead be put in charge of other projects.

Table 4 Correlates of Project Outcomes: Alternative Project Manager Records

Dependent variable is binary success (0,1) in all specifications					
Explanatory variables:	(1)	(2)	(3)	(4)	(5)
	Full Baseline	Asia Baseline	Full Alternative #1	Asia Alternative #1	Full Alternative #2
CPIA rating	0.127*** (3.65)	0.114 (1.52)	0.132*** (3.74)	0.121 (1.56)	0.132*** (3.80)
—ADB interaction	−0.0562 (−0.44)	−0.0528 (−0.41)	−0.0578 (−0.44)	−0.0573 (−0.43)	−0.0589 (−0.45)
Real GDP per capita growth	0.964* (1.82)	1.411* (1.76)	1.112*** (2.10)	1.590*** (2.02)	1.022* (1.92)
—ADB interaction	1.819 (0.82)	0.962 (0.50)	2.037 (0.83)	1.115 (0.51)	2.157 (0.89)
Freedom House rating	−0.00631 (−1.20)	−0.00708 (−0.89)	−0.00619 (−1.16)	−0.00707 (−0.90)	−0.00683 (−1.29)
—ADB interaction	−0.00342 (−0.21)	−0.00121 (−0.08)	−0.00325 (−0.19)	−0.00105 (−0.07)	−0.00266 (−0.15)
Dummy for PAR/PPER evaluations (d)	−0.0459 (−1.41)	0.0400 (0.78)	−0.0471 (−1.43)	0.0403 (0.76)	−0.0435 (−1.31)
—ADB interaction (d)	−0.296*** (−2.18)	−0.389*** (−2.57)	−0.256* (−1.93)	−0.352*** (−2.32)	−0.254* (−1.95)
Dummy for ADB PVR evaluations (d)	−0.284*** (−2.24)	−0.243*** (−2.20)	−0.278*** (−2.29)	−0.239*** (−2.25)	−0.272*** (−2.28)
Dummy for investment lending instrument (d)	0.141 (1.51)	−0.00380 (−0.03)	0.139 (1.48)	−0.00352 (−0.02)	0.141 (1.52)
					0.0439 (0.82)
					−0.000763 (−0.05)
					0.0439 (0.82)
					−0.351*** (−2.32)
					−0.234*** (−2.23)
					0.00493 (0.03)

Dependent variable is binary success (0,1) in all specifications

Explanatory variables:	(1)	(2)	(3)	(4)	(5)	(6)
	Full Baseline	Asia Baseline	Full Alternative #1	Asia Alternative #1	Full Alternative #2	Asia Alternative #2
—ADB interaction	0.0103 (0.09)	0.113 (0.85)	0.0338 (0.29)	0.131 (0.95)	0.0258 (0.23)	0.118 (0.85)
Log(total commitment)	0.0273** (2.28)	0.0258 (1.05)	0.0285** (2.33)	0.0244 (0.96)	0.0265** (2.20)	0.0253 (1.01)
—ADB interaction	−0.00264 (−0.08)	−0.00480 (−0.14)	0.00385 (0.10)	0.00340 (0.09)	0.00456 (0.13)	0.00140 (0.04)
Planned project length	−0.00153* (−1.84)	−0.00321*** (−2.74)	−0.00151* (−1.81)	−0.00320*** (−2.68)	−0.00151* (−1.82)	−0.00328*** (−2.70)
—ADB interaction	0.000741 (0.28)	0.00254 (1.04)	0.00140 (0.56)	0.00310 (1.29)	0.00147 (0.58)	0.00324 (1.32)
Effectiveness delay	0.000580 (0.17)	0.0129 (1.42)	0.000543 (0.16)	0.0130 (1.46)	0.000533 (0.16)	0.0134 (1.54)
—ADB interaction	−0.00506 (−0.77)	−0.0167 (−1.52)	−0.00551 (−0.82)	−0.0173 (−1.56)	−0.00516 (−0.78)	−0.0174 (−1.59)
Implementation delay	−0.00161** (−2.04)	−0.00146 (−1.03)	−0.00162** (−2.05)	−0.00132 (−0.97)	−0.00158** (−2.00)	−0.00135 (−0.97)
—ADB interaction	0.000959 (0.50)	0.000908 (0.38)	0.00120 (0.58)	0.000962 (0.38)	0.00127 (0.63)	0.00108 (0.44)
Log(additional funding)	0.227*** (4.84)	0.186*** (2.92)	0.224*** (4.91)	0.185*** (3.08)	0.223*** (4.87)	0.189*** (3.12)

Table 4 (continued)

Dependent variable is binary success (0,1) in all specifications					
Explanatory variables:	(1)	(2)	(3)	(4)	(5)
	Full Baseline	Asia Baseline	Full Alternative #1	Asia Alternative #1	Full Alternative #2
—ADB interaction	0.158** (1.97)	0.141 (1.97)	0.175** (2.27)	0.158* (1.81)	0.169** (2.21)
Project manager track record	0.239*** (3.79)	0.132 (1.31)	0.123*** (2.74)	0.0591 (2.74)	0.197*** (3.55)
—ADB interaction	0.175 (1.31)	0.221 (1.52)	−0.0262 (−0.20)	0.0239 (−0.20)	−0.0869 (−0.50)
Project manager turnover	−0.167** (−2.00)	−0.102 (−1.05)	−0.168** (−2.01)	−0.117 (−1.19)	−0.174** (−2.06)
—ADB interaction	0.0103 (0.07)	−0.0320 (−0.25)	−0.00873 (−0.06)	−0.00873 (−0.28)	0.00444 (−0.28)
Negative rating in first half (d)	−0.688*** (−22.86)	−0.736*** (−16.53)	−0.736*** (−23.13)	−0.741*** (−23.13)	−0.691*** (−22.86)
—ADB interaction (d)	0.258*** (18.66)	0.212*** (9.48)	0.261*** (18.16)	0.217*** (9.41)	0.259*** (18.32)
Observations	1919	851	1919	851	1919
					851

Table reports marginal effects from probit regression. “(d)” indicates discrete change of dummy variable from 0 to 1. T statistics based on heteroskedasticity-consistent standard errors clustered at the country level are reported in parentheses

*** (**) (*) denote significance at the 1 (5) (10) percent level. All regressions include sector and evaluation year fixed effects and their interactions with the ADB dummy

Table 5 Difference in Project Manager Track Record between the First and Second Half of Project

First half rating	Final project rating	(1) WB	(2) ADB
Non-success	Non-success	-0.006 (0.018) 275	0.086 (0.158) 10
Non-success	Success	0.045 (0.052) 27	0.060 (0.086) 17
Success	Non-success	0.019 (0.015) 356	-0.029 (0.040) 97
Success	Success	-0.007 (0.006) 1749	0.010 (0.025) 177

Table reports mean change in average project manager track records in the first half and second half of a project's life. Standard errors are reported in parentheses. Number of observations are reported below standard errors

***(**) (*) denote significance at the 1 (5) (10) percent level

the mean change in project track record in four groups of projects: those with negative ratings in the first half and upon completion; those with negative first half ratings and positive final ratings; those with positive first half ratings but negative final ratings; and those with positive ratings first half and final ratings.

The results are summarized in Table 5, and are somewhat mixed. In the ADB, the first two rows suggest that projects with negative first half ratings do on average experience improvements in project manager quality in the second half relative to the first – the average change in project manager track record is positive. This would be consistent with turnover resulting in on average “better” project managers being assigned to projects experiencing difficulty in their first half. In both the WB and ADB, projects that turned from a negative interim rating to a positive final rating had on average an improvement in project manager track record. However, none of these changes in project manager track record are statistically significant, although — particularly in the ADB — this may be due to the very small sample size. Turning to initially successful projects, in the WB we find, counterintuitively, that project manager track record actually improves in projects that nevertheless ended up with a negative final rating, and that project manager track record deteriorates in projects that have successful interim and final ratings. However, these effects are quite small when compared with the improvement in project manager track record observed in projects that turned around from non-success to success. Overall, this analysis does not suggest a clear pattern whereby “better” project managers are assigned to projects experiencing implementation problems — a potentially beneficial type of turnover.

5 Conclusions and policy implications

This paper has compared the country- and project-level correlates of success for a large sample of WB and ADB projects. The results on macro and micro correlates of project success that we find generally support those in the previous literature, particularly DKK. In terms of our institutional comparison, we cannot reject the null hypothesis that the size and significance of the relationship between most of these macro and micro correlates and project outcomes is the same for the WB and ADB. This suggests that the findings in DKK generalize to other large multilateral aid agencies, rather than reflecting particular institutional rules and norms of the WB.

Our findings with regard to the macro correlates of project success are similar to those of DKK. We find that country-level characteristics account for only 10–25 % of the variation in project outcomes, with the remaining variation occurring across projects within countries. Among country-level factors, we find that GDP growth and a good policy environment are positively correlated with project success. The significance of CPIA scores in particular helps to justify the WB's Performance Based Allocation system and the ADB's Asian Development Fund system, both of which use these scores to determine cross-country aid allocation. One objective of these aid allocation rules is to increase the likelihood that aid projects will achieve their intended outcomes, and the fact that successful project outcome ratings are more likely in countries with better policy performance supports this approach. However, the fact that most of the variation in project outcomes occurs within — rather than between — countries suggests that country-level aid allocation rules can only go so far — there is scope to further improve aid effectiveness by paying closer attention to project-level factors.

Our finding that for projects in Asia, civil liberties and political diversity at the country level are not positively correlated with project outcomes, differs from both DKK and Isham et al. (1997). The implication, that some one-party polities in Asia may not have less project implementation capacity, is not surprising given the countries in question — in particular, China and Vietnam — but the broader policy implications of such a finding are limited. We would not suggest that civil liberties and political freedoms are orthogonal to project performance, but simply that certain centralized countries with strong bureaucracies and stronger control over land and local governments have managed to implement projects successfully despite lower scores in these areas.

Our main findings with regard to project-level correlates are worth reiterating. We find that the difference between actual and initially-planned funding is positively correlated with project outcomes, likely reflecting the fact that projects that are not doing well are closed early. Intuitively, we find evidence that projects taking longer to implement are also more likely to be rated poorly. To the extent that project length is a proxy for project complexity (in addition to project size), this suggests that project complexity may be associated with lower success — projects should not try to do too much. It also suggests that extending projects to attempt to achieve goals in spite of hitches during implementation may not

always be successful, although there is also the possibility that outcomes could be even worse if bad projects are closed without trying to solve remaining problems. We find some evidence that the delay between project approval and the meeting of conditions for loan effectiveness on a project signals better project outcomes. As discussed above, a potential channel for this effect is that some countries and executing agencies do not meet conditions until detailed project designs have been completed and all procurement packages are prepared; therefore, the observed “delay” may in fact indicate concern to reduce commitment charges later, as well as increased care and attention in project planning. This reinforces the lesson that it helps if detailed project designs and procurement packages are prepared prior to project implementation, i.e., that project readiness is improved.

Leading indicators of project success, in the form of negative project ratings by staff during the first half of a project, are very strongly correlated with eventual negative project outcome ratings. This suggests that despite these early warning indicators, it is difficult to turn around projects experiencing difficulty during their implementation period. This finding implies that some combination of (a) better initial project design and readiness, (b) greater efforts to turn around problem projects, and (c) a greater willingness to cut short failing projects, could contribute to better overall project outcomes.

Finally, and perhaps most importantly, our finding that the track record of the project manager is a strong correlate of eventual project outcomes implies that aid organizations could improve aid effectiveness by devoting more effort to project manager selection, training, screening, and supervision. This could be complemented by giving more project management responsibilities to project managers with good track records on past projects. While this finding is in some ways an intuitive one – project managers matter – it is one that is often overlooked in discussions of the importance of institutional rules and implementation agency quality.

Generally, the importance of project-level correlates of success and the low degree of variation in outcome ratings explained by country-level correlates suggest that there may be scope for improving project quality through greater attention to project-level variables in addition to country-characteristics. This is consistent with the discussion in DKK and Gelb (2010), who note that finding ways to allocate aid on project-level indicators is important as a way to provide countries with greater incentives to ensure project success, as well as to scale up successful projects in countries that have worse policy and institutional environments. Our findings add weight to this conclusion by highlighting that the project-level correlates of project success are largely the same in both WB and ADB institutional settings, implying that these findings are more broadly relevant. Finally, it is worth acknowledging that, although we have identified a number of strong project-level correlates of project outcomes, much of the within-country variation in project outcomes remains unaccounted for in our analysis. This points to the importance of further work to understand the sources of this variation, for example, by systematically measuring the contribution to project success of project implementing agencies within recipient governments.

Acknowledgments The authors are grateful to Adele Casorla for help compiling the data of the Independent Evaluation Department of the Asian Development Bank, and to Adele Casorla, Luis Serven, Ganesh Rauniyar, Vinod Thomas, Jiro Tominaga, and Hans Van Rijn for helpful comments on earlier drafts, and to the Editor and two anonymous referees for very helpful feedback. The views expressed here are the authors' and do not reflect the official views of the World Bank, the Asian Development Bank, their Executive Directors, or the countries they represent.

Appendix

Table 6 Descriptive Statistics

	World Bank			Asian Development Bank		
	Mean	St. Dev.	Obs.	Mean	St. Dev.	Obs.
Success	0.74	0.44	3797	0.65	0.48	1322
CPIA rating	3.44	0.50	3797	3.48	0.51	1322
Real GDP per capita growth	0.03	0.03	3797	0.04	0.03	1322
Freedom House rating	6.08	2.99	3797	4.91	2.62	1322
Dummy for PAR/PPER evaluations	0.27	0.44	3797	0.48	0.50	1322
Dummy for ADB PVR evaluations	-	-	3797	0.19	0.39	1322
Log(total commitment)	3.99	1.24	3797	4.07	1.01	1322
Planned project length	57.38	25.10	3797	51.33	18.24	1322
Effectiveness delay	6.54	4.61	3797	6.51	4.35	1322
Implementation delay	16.33	15.92	3797	23.23	23.87	1322
Log(additional funding)	-0.15	0.41	3797	-0.24	0.46	1322
Project manager track record	0.72	0.28	2373	0.68	0.28	367
Project manager track record (alt. #1)	0.71	0.32	1591	0.66	0.32	329
Project manager track record (alt. #2)	0.72	0.29	1591	0.68	0.29	329
Project manager turnover	0.43	0.22	2373	0.63	0.27	367
Negative rating in first half	0.13	0.33	2373	0.09	0.28	367

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